# A Literature Review on the Accuracy Assessment and Bias Correction of Gridded Climate Products for Nepal

## 1.0 Introduction: The Critical Need for Reliable Climate Data in Nepal

### 1.1 The Challenge of a Data-Scarce, High-Mountain Environment

Accurate and reliable data on precipitation and temperature are strategically vital for Nepal. These datasets form the foundation for effective water resource management, agricultural planning, and hydropower generation (Gautam & Pradhananga, 2024). Furthermore, they are indispensable for understanding and mitigating the impacts of climate change, which manifest as increasingly frequent and intense floods and droughts (Mishra, 2020). The pronounced variability of the monsoon system, which supplies the majority of the region's freshwater, underscores the critical need for high-quality climate information to safeguard lives and livelihoods (Bohlinger, 2018).

The primary challenge in developing such datasets for Nepal is the inherent scarcity of ground-based observations. Conventional climate monitoring relies on in-situ rain gauges, but the network in Nepal is sparse and unevenly distributed, particularly in remote, high-altitude regions of the Himalayas (Ali et al., 2024; Bohlinger, 2018). This data scarcity is a principal driver for the adoption of gridded climate products (Oukaddour et al., 2015). However, these products face their own significant hurdles in the region. The extreme topography of the Himalayas profoundly influences local climate through complex interactions between terrain and atmosphere (Chen et al., 2021). This creates strong precipitation gradients due to orographic uplift on windward slopes and rain-shadow effects on leeward sides, as well as significant temperature variations with elevation (lapse rate effects). Most gridded products, particularly those with coarser spatial resolutions, struggle to realistically capture these fine-scale, topography-driven climate features (Sun, 2017).

### 1.2 The Role of Gridded Precipitation Products (GPPs)

To overcome the limitations of sparse gauge networks, Gridded Precipitation Products (GPPs) have emerged as a viable and essential alternative. By leveraging satellite observations, climate model reanalysis, and interpolation of existing gauge data, GPPs provide continuous spatiotemporal coverage, offering a complete picture of climate patterns where ground data is missing (Sun, 2017). They are a crucial resource for regional climate monitoring, trend analysis, and as inputs for hydrological and ecological models.

However, the process of generating these products introduces its own errors and biases. Therefore, before these datasets can be confidently used for local impact studies, their reliability must be rigorously evaluated. This literature review serves a dual purpose. First, it surveys the methodologies and principal findings from studies that have assessed the accuracy

of various GPPs against ground observations in Nepal and the broader Himalayan region. Second, it reviews the necessity and application of bias correction techniques, which are critical post-processing steps to improve the usability of GPPs for scientific research and operational decision-making (Shrestha, 2017).

## 2.0 A Taxonomy of Gridded Climate Products

### 2.1 Overview and Categorization

Gridded climate products are not a monolithic category; they are derived from a diverse array of sources and methodologies, each with inherent strengths and limitations (Sun, 2017). Understanding these distinctions is fundamental to selecting an appropriate product and interpreting evaluation results. A consensus emerges from the literature that the primary products available can be classified into four main categories based on their data sources and generation process: gauge-based, satellite-based, reanalysis, and merged products.

- **Gauge-Based Gridded Products**
    - **Definition:** These products are created by spatially interpolating data collected from in-situ rain gauge stations. The core assumption is that rainfall values are spatially correlated, allowing for the estimation of precipitation at unmeasured locations (Pai et al., 2014). Common interpolation methods discussed in the literature include Inverse Distance Weighting (IDW), Kriging, and the Shepard algorithm (Tostes et al., 2017; Pai et al., 2014).
    - **Examples:** APHRODITE (Yatagai et al., 2012), CRU-TS (Harris et al., 2014), GPCC (Schneider et al., 2013), UDEL (Willmott & Matsuura, 2001), and IMD4, a high-resolution (0.25°) daily gridded product for India developed from over 6900 stations (Pai et al., 2014).
- **Satellite-Based Products**
    - **Definition:** These products estimate precipitation by using data from sensors aboard satellites. Algorithms interpret measurements of infrared (IR) brightness temperature or passive microwave (PMW) radiation to infer rainfall rates, providing excellent coverage over oceans and sparsely populated land areas (Dhmane et al., 2023; Sun, 2017).
    - **Examples:** TRMM/TMPA (noting that some versions, like 3B43, are merged products that incorporate gauge data) (Huffman et al., 2007; Kanda et al., 2020), IMERG (including its near-real-time 'Early' and 'Late' runs and the research-grade 'Final' run: GPM-E, GPM-L, GPM-F) (Dhmane et al., 2023), and PERSIANN-CDR (Ashouri et al., 2015).

- **Reanalysis Products**
    - **Definition:** Reanalysis is a scientific method that synthesizes historical weather observations from various sources (e.g., ground stations, weather balloons, satellites) and assimilates them into a modern weather forecast model. This process generates a comprehensive, spatially complete, and physically consistent gridded dataset representing the state of the atmosphere over long periods (Sun, 2017).
    - **Examples:** ERA-Interim (Dee et al., 2011), ERA5 (Hersbach et al., 2020), ERA5-Land (Hersbach et al., 2020), and MERRA-2 (Gelaro et al., 2017).
- **Merged and Hybrid Products**
    - **Definition:** These products optimally combine data from multiple sources—typically gauge, satellite, and/or reanalysis—to leverage the strengths of each. For instance, satellite data provide broad coverage, while gauge data are used to correct biases and improve accuracy over land (Sun, 2017).
    - **Examples:** MSWEPv2.8 (Beck et al., 2019) and the Princeton Global Forcings Dataset (PGF) (Sheffield et al., 2006).

This categorization provides a framework for understanding the fundamental differences between products. The next section will outline the methods used to validate their performance against ground-truth observations.

## 3.0 Methodologies for Accuracy Assessment

### 3.1 Evaluation Frameworks

A robust evaluation framework is essential for quantifying the reliability of GPPs and determining their suitability for specific applications (Kanda et al., 2020). Before a gridded dataset can be used to drive a hydrological model or assess long-term climate trends, its systematic errors and uncertainties must be understood. The literature employs several common approaches to validate these products against ground-based observations.

- **Point-to-Pixel / Station-to-Grid Comparison:** This is the most direct validation method. Data from an individual rain gauge station are compared directly against the precipitation values from the corresponding grid cell in which the station is located. This approach is widely adopted because it avoids introducing additional errors that could arise from spatially interpolating the already sparse station data to create a gridded reference field (Chen et al., 2021; Dhmane et al., 2023).

- **Evaluation by Climate Zone and Altitude:** Recognizing that product performance can vary significantly with topography, many studies segment their analysis by geographic or climatic zones. For instance, evaluations in the Himalayas have divided the study area into distinct zones like the Lower Himalaya, Greater Himalaya, and Karakoram Himalaya (Kanda et al., 2020). Similarly, studies in other mountainous regions have assessed performance across plains, piedmonts, and mountains, revealing that a product's biases are often elevation-dependent (Oukaddour et al., 2015).
- **Evaluation by Precipitation Intensity:** The accuracy of a GPP is often not uniform across different types of rain events. To assess this, precipitation events are classified into different intensity categories (e.g., light, moderate, heavy, and extreme). The performance of the product is then evaluated for each class separately, revealing whether a product accurately captures drizzling rain but fails to represent torrential downpours, or vice versa (Dhmane et al., 2023; Gautam & Pradhananga, 2024).

### 3.2 Key Performance Metrics

Across these evaluation frameworks, a standard set of statistical metrics is used to quantify performance. These can be broadly divided into regression/error metrics, which measure the magnitude of errors, and categorical metrics, which assess a product's ability to correctly detect rain versus no-rain events (Gautam & Pradhananga, 2024).

| Metric Category | Metrics and Descriptions |
|---|---|
| **Regression & Error Metrics** | **Correlation Coefficient (r or CC):** Measures the linear relationship between estimated and observed values. <br> **Root Mean Square Error (RMSE):** Measures the average magnitude of errors, giving higher weight to large errors. <br> **Mean Absolute Error (MAE):** Measures the average magnitude of errors without squaring them, making it less sensitive to outliers. <br> **Bias / Percent Bias (PBIAS):** Indicates the average tendency of the estimated values to be larger or smaller than the observed values. <br> **Kling-Gupta Efficiency (KGE):** A composite metric that evaluates correlation, bias, and variability. |
| **Categorical / Binary Metrics** | **Probability of Detection (POD) / Recall:** Measures the fraction of observed rain events that were correctly forecasted. <br> **False Alarm Ratio (FAR):** Measures the fraction of forecasted rain events that did not occur. <br> **Percent Correct (PC) / Accuracy:** Measures the overall fraction of correct forecasts (both rain and no-rain). <br> **Frequency Bias (BIAS Score):** Compares the frequency of forecasted events to the frequency of observed events. |

These evaluation frameworks and metrics provide the quantitative basis for the findings synthesized in the following section.

## 4.0 Performance of Gridded Products in Nepal and the Himalayas: A Synthesis of Findings

### 4.1 Common Biases and Performance Patterns

A consistent picture emerges from the literature regarding the performance of GPPs in Nepal and similar high-mountain regions. While most products successfully capture the broad spatial patterns and seasonal cycles of precipitation and temperature, they also exhibit significant and systematic biases when compared to ground observations (Chen et al., 2021). These errors vary by product type, season, elevation, and event intensity, underscoring the complexity of modeling climate in such environments.

- **Systematic Over/Underestimation** Many reanalysis products, including ERA5 and ERA5-Land, systematically overestimate precipitation in regions with complex topography. This tendency has been documented in the Central Himalayas (Chen et al., 2021), the semi-arid Tensift basin in Morocco (Oukaddour et al., 2015), and northwestern Morocco (Dhmane et al., 2023), with the bias often increasing with altitude. Conversely, some satellite-based products like TRMM tend to underestimate precipitation (Sun, 2017). For temperature, some global climate models (CMIP6) exhibit a substantial cold bias in the Himalayan region (Mishra, 2020).
- **Challenges with Orographic Precipitation** A prevalent weakness across nearly all GPPs is the difficulty in accurately representing orographic precipitation. This is a consistent finding, whether evaluating reanalysis products in the Himalayas (Chen et al., 2021), satellite products in South Asia (Sun, 2017), or even gauge-based products like APHRODITE, which still struggle with orographic representation despite being informed by a dense station network (Pai et al., 2014). The models struggle to capture the sharp precipitation gradients found on the windward and leeward sides of mountain ranges.
- **Performance Variation by Season and Intensity** Product accuracy is not static throughout the year. For instance, in a semi-arid Moroccan basin, products performed better during the dry summer months compared to the wet season (Dhmane et al., 2023), and in the Himalayas, reanalysis accuracy varies significantly between seasons (Li et al., 2022). Furthermore, accuracy often varies with event intensity. For precipitation, products are typically most accurate for light rain, with performance degrading for moderate, heavy, and extreme events. For instance, some models over-forecast the frequency of light and moderate rain while under-forecasting the occurrence of heavy rain (Gautam & Pradhananga, 2024; Dhmane et al., 2023).

- **Impact of Spatial Resolution** Spatial resolution is a critical factor influencing product performance. A consensus emerges that higher-resolution datasets, such as the High Asia Refined analysis (HAR), are better able to represent precipitation patterns realistically by resolving finer topographic features (Chen et al., 2021). In contrast, coarse-resolution products like MERRA-2 often fail to capture important local climate features and may misrepresent seasonality (Oukaddour et al., 2015).

### 4.2 Ranking and Suitability

Based on comprehensive evaluations in the Northwestern Himalayas, specific products have been identified as more reliable for the region (Kanda et al., 2020). The reanalysis product ERA-Interim, the gauge-based product GPCC, and the merged satellite-gauge product TRMM/TMPA performed well for precipitation. For temperature, the gauge-based CRU-TS and the ERA-Interim reanalysis provided the most reliable estimates (Kanda et al., 2020).

Despite these relative rankings, the absolute magnitude of errors remains a significant concern. The same study reported Mean Absolute Error (MAE) for wintertime monthly precipitation ranging from 13.5 mm/month to as high as 150.7 mm/month (Kanda et al., 2020). These large errors reinforce a critical consensus in the literature: bias correction is an absolute necessity before these products can be reliably used for local-scale hydrological modeling or climate impact assessment.

## 5.0 The Imperative and Practice of Bias Correction

### 5.1 Rationale for Bias Correction

Bias correction is a critical and non-negotiable step in the workflow of applying GPPs to real-world problems. Raw outputs from climate models and reanalysis systems contain systematic errors that stem from model physics, parameterizations, and resolution limitations (Shrestha, 2017). If these raw, biased data are used directly as inputs for impact models—such as a hydrological model used to simulate river discharge—the resulting outputs will be unreliable and potentially misleading. A study of the Kaligandaki River basin in Nepal demonstrated this point clearly: when a hydrological model was forced with raw RCM data, the simulated river discharge was grossly overestimated. However, after the RCM data were bias-corrected, the model's performance improved dramatically, closely matching observed discharge (Shrestha, 2017). This illustrates that bias correction is essential to bridge the gap between gridded climate estimates and their practical application.

### 5.2 A Spectrum of Bias Correction Techniques

A variety of bias correction techniques exist, ranging from simple scaling adjustments to more complex methods that align the full statistical distribution of the data. The literature from Nepal and similar regions has explored several of these approaches.

- **Simple Scaling Methods**
    - **Linear Scaling (LS):** This straightforward method adjusts the mean of the simulated data to match the mean of the observed data. The correction is typically applied on a monthly basis using an additive factor for temperature or a multiplicative factor for precipitation. This approach is also referred to as correction by **monthly correction factors** (Shrestha, 2017).
- **Distribution-Based Methods**
    - **Quantile Mapping (QM):** Also known as **Empirical Quantile Mapping (EQM)**, this is a more sophisticated technique. Instead of only correcting the mean, QM adjusts the entire probability distribution of the modeled data. It achieves this by mapping each quantile of the modeled Cumulative Distribution Function (CDF) to the corresponding quantile of the observed CDF, ensuring that the corrected data has a statistical distribution that closely matches the observations (Mishra, 2020; Shrestha, 2017).
- **Parametric and Non-Parametric Transformations**
    - In addition to the primary methods above, researchers have tested a suite of other transformation schemes. These include parametric methods such as **power** and **linear** transformations, and non-parametric methods like **robust empirical quantiles** and **smoothing spline**, which offer different ways to model the relationship between observed and simulated data (Gautam & Pradhananga, 2024).
- **Machine Learning Approaches**
    - **Artificial Neural Network (ANN):** This advanced method uses a machine learning model to learn the complex, non-linear relationships between the raw gridded data and the observed ground-truth data. Once trained, the ANN can be used to correct the biases in the gridded product (Dhmane et al., 2023).

### 5.3 Application-Specific Considerations and Effectiveness

Studies assessing these techniques have shown that more complex methods are often, but not always, superior. A critical nuance identified by multiple studies is that the choice of method depends heavily on the temporal scale of the analysis.

- For correcting **daily** weather forecasts from the Weather Research and Forecasting (WRF) model in Nepal's Babai River Basin, a detailed comparison identified **Empirical Quantile Mapping (EQM)** as the most effective scheme for both precipitation and temperature (Gautam & Pradhananga, 2024).
- However, a hydrological study in the Kaligandaki basin revealed that when the analysis was conducted at a **monthly temporal resolution**, the simple **Linear Scaling** method performed almost identically to the more complex Quantile Mapping technique for correcting RCM outputs for hydrological modeling (Shrestha, 2017).
- In a semi-arid, mountainous region in Morocco, the **Artificial Neural Network (ANN)** approach proved highly successful for correcting a satellite precipitation product, leading to a significant reduction in biases, RMSE, and MAE (Dhmane et al., 2023).

These findings highlight that while bias correction is essential, the choice of the optimal method is not universal. The decision depends heavily on the specific application, the temporal scale of the analysis (daily vs. monthly), and the climate variable being corrected. For some applications, simpler methods may be sufficient, while others may require the statistical rigor of more advanced techniques.

**6.0 Conclusion and Future Directions**

This review affirms a dual reality for climate science in data-scarce regions like Nepal. Gridded climate products are an indispensable tool, providing vital information where ground observations are lacking. Yet, their direct use is unadvisable due to significant inherent biases. Rigorous accuracy assessment and the application of appropriate bias correction are mandatory prerequisites for their use in any local-scale impact modeling.

A consensus emerges from the literature on the primary challenges: the persistent, fundamental difficulty of accurately capturing orographic precipitation effects in complex terrain, representing the intensity and frequency of extreme events, and the residual uncertainty that persists even after bias correction. While current methods can significantly improve the quality of gridded datasets, they are not a panacea.

Looking forward, research should focus on advancing the science of bias correction. There is a need to develop and test more sophisticated techniques that are robust under non-stationary climate conditions, as the statistical relationships between modeled and observed data may themselves change in a warming world (Teutschbein & Seibert, 2013). Furthermore, future methods should aim to better incorporate the physical processes driving precipitation and temperature in complex mountain environments, moving beyond purely statistical adjustments to process-informed corrections. Such advancements will be crucial for improving the reliability

of climate projections and supporting effective adaptation strategies in vulnerable regions like the Himalayas.

---

**References**


Ali, S., Bhutta, Z.A., Reboita, M.S., Goheer, M.A., Ebrahimi, S., Rozante, J.R. et al. (2024). A 5-km gridded product development of daily temperature and precipitation for Bangladesh, Nepal, and Pakistan from 1981 to 2016. *Geoscience Data Journal*, 11, 292–302.

Ashouri, H., Hsu, K. L., Sorooshian, S., Braithwaite, D. K., Knapp, K. R., Cecil, L. D., ... & Prat, O. P. (2015). PERSIANN-CDR: Daily 0.25-degree precipitation climate data record from multisatellite observations for hydrological and climate studies. *Bulletin of the American Meteorological Society*, 96(1), 69-83.

Beck, H. E., Wood, E. F., Pan, M., Fisher, C. K., Miralles, D. G., van Dijk, A. I., ... & Adler, R. F. (2019). MSWEP V2 global 3-hourly 0.1° precipitation: methodology and quantitative assessment. *Bulletin of the American Meteorological Society*, 100(3), 473-500.

Bohlinger, P. (2018). *Extreme Precipitation in Nepal: Trends and Key Processes*. PhD thesis, University of Bergen, Norway.

Chen, Y., Sharma, S., Zhou, X., Yang, K., Li, X., Niu, X., ... & Khadka, N. (2021). Spatial performance of multiple reanalysis precipitation datasets on the southern slope of central Himalaya. *Atmospheric Research*, 250, 105365.

Dee, D. P., Uppala, S. M., Simmons, A. J., Berrisford, P., Poli, P., Kobayashi, S., ... & Vitart, F. (2011). The ERA-Interim reanalysis: Configuration and performance of the data assimilation system. *Quarterly Journal of the Royal Meteorological Society*, 137(656), 553-597.

Dhmane, L. A., Moustadraf, J., Rachdane, M., Saidi, M. E., Benjmel, K., Amraoui, F., ... & Hadri, A. (2023). Spatiotemporal Assessment and Correction of Gridded Precipitation Products in North Western Morocco. *Atmosphere*, 14(8), 1239.

Gautam, D. K., & Pradhananga, R. D. (2024). Verification and Bias Correction of Rainfall and Temperature Forecasts over the Babai River Basin of Nepal. *Journal of Nepal Geological Society*.

Gelaro, R., McCarty, W., Suárez, M. J., Todling, R., Molod, A., Takacs, L., ... & Zhao, B. (2017). The modern-era retrospective analysis for research and applications, version 2 (MERRA-2). *Journal of Climate*, 30(14), 5419-5454.

Harris, I., Jones, P. D., Osborn, T. J., & Lister, D. H. (2014). Updated high-resolution grids of monthly climatic observations–the CRU TS3. 10 Dataset. *International Journal of Climatology*, 34(3), 623-642.

Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., ... & Simmons, A. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730), 1999-2049.

Huffman, G. J., Adler, R. F., Arkin, P., Chang, A., Ferraro, R., Gruber, A., ... & Rudolf, B. (1997). The Global Precipitation Climatology Project (GPCP) combined precipitation dataset. *Bulletin of the American Meteorological Society*, 78(1), 5-20.

Huffman, G. J., Adler, R. F., Bolvin, D. T., Gu, G., Nelkin, E. J., Bowman, K. P., ... & Wolff, D. B. (2007). The TRMM multisatellite precipitation analysis (TMPA): Quasi-global, multiyear, combined-sensor precipitation estimates at fine scales. *Journal of Hydrometeorology*, 8(1), 38-55.

Kanda, N., Negi, H. S., Rishi, M. S., & Kumar, A. (2020). Performance of various gridded temperature and precipitation datasets over Northwest Himalayan Region. *Environmental Research Communications*, 2(8), 085002.

Li, Y., Qin, X., Jia, Z., Lu, J., Wang, L., & Chen, J. (2022). Evaluation of CMFD and ERA5-Land precipitation and temperature in the Qilian Mountains. *Frontiers in Earth Science*, 10, 831415.

Mishra, V. (2020). Bias-corrected climate projections for South asia from Coupled Model Intercomparison Project-6. *Scientific Data*, 7(1), 328.

Oukaddour, A., Jarlan, L., Lo, M. M., Ezzahar, J., Ait Hssaine, B., & Chehbouni, A. (2015). A Comparative Analysis of Gridded Precipitation Products in a Semi-Arid and Mountainous Environment (Tensift Basin, Morocco). *Geomatics, Natural Hazards and Risk*. [Note: While the reference found in the context is a preprint/accepted manuscript from 2025, the key findings cited refer to the study conducted over the Tensift basin, which matches the provided content.]

Pai, D. S., Sridhar, L., Rajeevan, M., Sreejith, O. P., Satbhai, N. S., & Mukhopadhyay, B. (2014). Development of a new high spatial resolution (0.25° × 0.25°) long period (1901-2010) daily gridded rainfall data set over India and its comparison with existing data sets over the region. *Mausam*, 65(1), 1-18.

Schneider, U., Becker, A., Finger, P., Meyer-Christoffer, A., Rudolf, B., & Ziese, M. (2013). GPCC's new land surface precipitation climatology based on quality-controlled in-situ data and its role in quantifying the global water cycle. *Theoretical and Applied Climatology*, 115(1-2), 15-40.

Sheffield, J., Goteti, G., & Wood, E. F. (2006). Development of a 50-year high-resolution global dataset of meteorological forcings for land surface modeling. *Journal of Climate*, 19(13), 3088-3111.

Shrestha, M., Artan, G., Bajracharya, S. R., & Sharma, R. R. (2017). Are simple bias correction methods still useful for hydrological analysis in the Himalayas? *Meteorological Applications*, 24(3), 531-539.

Sun, Q., Miao, C., Duan, Q., Ashouri, H., Sorooshian, S., & Hsu, K. L. (2017). A review of global precipitation data sets: Data sources, estimation, and intercomparisons. *Reviews of Geophysics*, 56(1), 79-107.

Teutschbein, C., & Seibert, J. (2013). Is bias correction of regional climate model (RCM) simulations possible for non-stationary conditions? *Hydrology and Earth System Sciences*, 17(12), 5061-5077.

Tostes, J. O., de Oliveira, A. L., & Schaefer, C. E. G. R. (2017). Performance analysis of rainfall and air temperature products (gridded data) in the state of Acre. *Environmental Earth Sciences*, 76, 153.

Willmott, C. J., & Matsuura, K. (2001). Terrestrial air temperature and precipitation: monthly and annual time series (1950–1999). *Version 1.02*.

Yatagai, A., Kamiguchi, K., Arakawa, O., Hamada, A., Yasutomi, N., & Kitoh, A. (2012). APHRODITE: Constructing a long-term daily gridded precipitation dataset for Asia based on a dense network of rain gauges. *Bulletin of the American Meteorological Society*, 93(9), 1401-1415.